John Hancock SOFR IRS — Group Project Case

Topic 14 | Verified SEC terms vs. proposed modelling assumptions | One-page brief

A. CONFIRMED — SEC fund holdings (not hypothetical)

Term	Disclosed information
Fund / instrument	John Hancock Financial Opportunities Fund (BTO); centrally cleared USD interest rate swap
Position / notional	Fund pays fixed 3.356%; receives USD Compounded SOFR; notional USD 10,000,000
Payment frequency	Fixed: semi-annual | Floating: quarterly
Maturity	May 2028 (exact day NOT disclosed)
Reported fund value	+USD 275,600 on 30 Jun 2023; +USD 117,354 on 30 Jun 2026

B. ASSUMED — working contract for our modelling only

Assumed item	Working choice (not a verified trade term)
Model effective date	01 May 2023; chosen for a clean April/May data split, NOT the confirmed actual start date
Exact model maturity	01 May 2028 (modelled 5-year term); actual execution/trade date remains unknown
Floating / notional	0 bp spread above compounded SOFR; USD 10m constant; no principal exchange or amortisation
Conventions	ACT/360 both legs; US Gov. Securities business calendar; Modified Following; 2-business-day payment lag and SOFR lookback
Valuation	SOFR OIS curve for projection/discounting; assume zero initial upfront payment; initial model NPV need not equal zero

C. DATA SPLIT & TESTS

In-sample: market data dated before 01 May 2023 (last trading day: 28 Apr 2023). **Out-of-sample:** from 01 May 2023 onward; daily risk forecasts and realised model P&L for VaR/ES backtesting. **Separate validation:** compare modelled values with the fund's SEC-reported values on report dates.

Timing caveat: A 3.356% fixed coupon and May 2028 maturity are consistent with an approximately May 2023 five-year swap, but cannot identify the actual trade date. Our 01 May start is an explicit convenience assumption.

SEC PRIMARY SOURCES

[1] 2026 semiannual report, p.10 — DERIVATIVES / SWAPS
[2] 2023 semiannual report, p.10 — first reported swap value